(DEMO) From Natural Language to Live 5G API Calls: A CAPIF-Compliant, Multi-LLM Intent Orchestration Platform

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***Abstract*—We demonstrate an open-source platform that converts free-text operator intents into live CAMARA API calls. A two-phase large-language-model (LLM) planner selects the CAMARA service and builds a schema-grounded payload, and a three-layer deterministic validator (schema, telecom semantics, live CAPIF registry) gates every call. Quality of Service (QoS) requests are authorized by the PCF of a containerized Open5GS 5G core with UERANSIM-emulated UEs. A live request takes 2.2–4.8 s with GPT-4o-mini or Claude Haiku 4.5 (6.9–8.2 s with Gemini), 78–99% of it LLM planning. On 300 unique bilingual intents, the LLM planners reach an Intent-to-API Workflow Success Rate (IWSR) of 95.3–97.3%, versus 83.3% for a keyword baseline and 16.3% without schema grounding. Code, testbed and data: https://github.com/doadurak/mini-telco-platform.**

***Index Terms*—CAPIF, CAMARA, intent-based networking, LLM orchestration, 5G API exposure, Open5GS**

I. Introduction

Zero-touch Beyond-5G management requires orchestration layers that translate high-level intents into standards-compliant API invocations [1]. 3GPP CAPIF [2] and GSMA CAMARA [3] standardize how verticals discover and consume network capabilities, but closing the intent-to-API gap on a *live* exposure stack remains open. OpenCAPIF [4] provides exposure infrastructure without NL planning; Turkovic et al. [5] ran live CAMARA QoD with hand-written payloads; Maatouk et al. [6] identify hallucination as the key risk of telecom LLMs, and Liu et al. [7] report 40% JSON-schema compliance for ungrounded GPT-4. Our platform combines live CAPIF onboarding, schema-grounded multi-LLM planning, a three-layer validator, and CAMARA execution down to a 3GPP PCF, reproducible from a single script.

II. Platform Architecture

Planning: A FastAPI backend exposes `POST /orchestrate` (Fig. 1). Phase 1 selects one of four CAMARA services (QoD v1.1, QoS Profiles, QoS Provisioning, Location Retrieval v0.5) from a schema-grounded catalog; Phase 2 populates the payload with the operation's OpenAPI schema injected [8]. Intent-level QoS classes (e.g. `QOS_E_STREAMING`) are mapped by target bit rate to the operator profiles the network enforces (`QOS_E/S/M/L`).

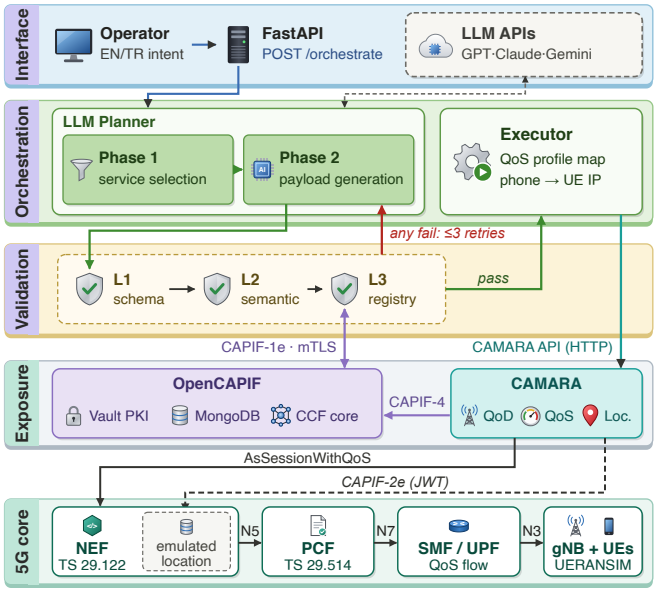


Fig. 1. Platform architecture. Any L1–L3 failure triggers at most three payload corrections; CAPIF-1e runs over mTLS, provider calls over HTTP.

Validation: L1 checks the payload against its Pydantic schema (pass rate: Schema Compliance Rate, SCR). L2 enforces telecom rules (E.164 numbers, QoS profiles that map to an operator profile, durations of 1–7200 s, unicast IPv4); its pass rate is the Semantic Validity Rate (SVR). L3 queries the live CAPIF registry to confirm that the selected service is published; a wrong-but-registered service passes and shows up as an IWSR miss against ground truth.

Execution: For QoS, the adapter resolves the phone number to the UE's IP from the 5GC subscriber database; the CAMARA provider forwards an AsSessionWithQoS subscription to the NEF, which creates a PCF application session bound to the UE's PDU session. Location uses the NEF Monitoring Event API with emulated core data.

III. Live Demonstration

The full stack (OpenCAPIF, four CAMARA/NEF providers, Open5GS, a UERANSIM gNB with three UEs) runs on

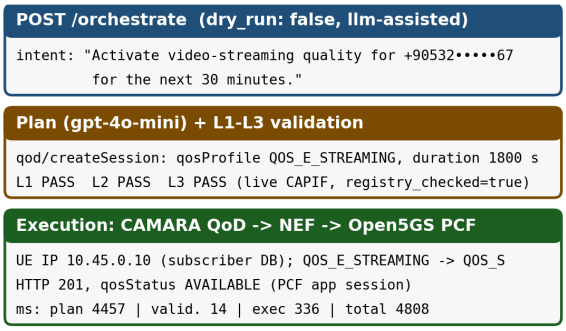


Fig. 2. Recorded live response (S1, GPT-4o-mini, `dry_run:false`), rendered from the stored JSON in the repository (MSISDN masked).

TABLE I
300-ENTRY OFFLINE EVALUATION (ONE RUN PER INTENT)

Planner	IWSR	SCR	SVR	HR	SS
Keyword baseline	83.3%	100.0%	100.0%	16.7%	0.962
GPT-4o-mini	**97.3%**	100.0%	99.0%	**2.3%**	**0.975**
Claude Haiku 4.5	95.3%	99.3%	**99.3%**	4.0%	0.927
Gemini 2.5 Flash	96.0%	100.0%	98.3%	3.0%	0.945
Ablations (GPT-4o-mini, no feedback loop)					
Single LLM call	77.3%	78.0%	77.3%	3.3%	0.949
No schema in Ph. 2	16.3%	16.7%	16.7%	4.0%	0.947

one 16 GB host via `start.sh`; all scenarios run with `dry_run:false` and are versioned in the repository. **S1 (QoD):** *"Activate video-streaming quality for ⟨MSISDN⟩ for the next 30 minutes"* yields `QOS_E_STREAMING`, 1800 s, enforced as `QOS_S`; the PCF accepts the session and QoD returns HTTP 201, `AVAILABLE` (Fig. 2). **S2 (Location):** the NEF returns a WGS-84 area (HTTP 200). **S3 (ambiguity):** dataset entry *"Provide network quality improvement for device ⟨MSISDN⟩ for a long time"* (ground truth: QoD). In the benchmark GPT-4o-mini chose QoD and Claude and Gemini QoS Provisioning; live, all three chose QoS Provisioning (PCF: `ACTIVE`). Every plan passes L1–L3 and executes: the validator guarantees well-formed, registered calls, not intent correctness. **S4 (backend switch):** after each `POST /config/provider` switch, S1 yields identical CAMARA payloads from all three LLMs. Validation takes 13–27 ms and execution 0.03–0.6 s. UERANSIM lacks NGAP PDU Session Resource Modify, so the dedicated QoS flow is authorized by the core but not enforced on the emulated radio. *Scalability and LLM APIs:* with the keyword planner and live L3, one backend process sustains ≈160 req/s (p95 156 ms, 20 concurrent clients); with GPT-4o-mini, latency stays at ≈2.6 s up to 10 concurrent requests while throughput grows linearly (0.34 to 3.3 req/s): the LLM API, not the platform, is the bottleneck. In `auto` mode an LLM error falls back to the keyword planner; since intents (with MSISDNs) leave the testbed, production use needs pseudonymization or a self-hosted model.

IV. EVALUATION

The dataset holds 300 unique intents (152 EN / 148 TR; 96 easy, 94 medium, 110 hard): 275 across the four services and 25 ambiguous edge cases. We derived the ground truth (service, operation, payload checks) from the CAMARA OpenAPI constraints and NEF test fixtures; it supersedes a 60-entry pilot (`datasets/legacy_pilot_60/`).[1] IWSR counts plans with the correct service that pass validation; HR is the share of wrong service selections; SS is $1 - \sigma$ of IWSR across difficulty strata, not run-to-run repeatability (temperature 0.1, so ambiguous choices can vary, cf. S3). The benchmark makes no provider calls; all 1,200 recorded plans pass the live L3 check. The keyword rules were written against the pilot (baseline 100%); on the 240 held-out entries the baseline drops to 79.2% and the LLMs reach 95.4–97.1%. LLM failures (8–14 per model) are mostly QoD vs. QoS Provisioning confusions as in S3; hard IWSR is 87–95% vs. 99–100% for easy and medium. *Ablations:* without the operation's schema and field rules in Phase 2, GPT-4o-mini invents field names and SCR falls to 16.7% (cf. 40% for ungrounded GPT-4 in [7]); a single-call planner reaches 77.3%, versus 96.7% for our two-phase planner without feedback. *Fault injection* (Table II): each fault is caught by its target layer; feedback repairs L1/L2 faults in one iteration but cannot fix an unpublished or wrong-but-registered service, since it only rewrites the payload.

TABLE II
FAULT INJECTION INTO CORRECT GPT-4O-MINI PLANS (LIVE L3)

Layer	Injected fault	Caught	Repaired	Feedback
L1	missing field, wrong type	40/40	40/40	1.3 s (1 it.)
L2	E.164, duration, profile, IPv4	80/80	79/80	1.3 s (1 it.)
L3	service not published	20/20	0/20	3.8 s (3 it.)

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[1]The first 60 LLM results per model are reused from the pilot; the Claude pilot predates three bug fixes, and a symptom scan found no affected entry. Eleven multi-intent entries were re-run after adding multi-step plans (flagged in the result files).